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Addendum to MISO DPP 2017 August Wisconsin Area

Phase 3 System Impact Study Report

J732, J818, J831, J850, J855, J864,

J870, J871, J878, J886, and J947

Version 1.1

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1.0 EXECUTIVE SUMMARY

This report is an addendum to MISO DPP 2017 August Wisconsin Area Phase 3 System Impact Study (SIS) report issued on 08/20/2019. It contains updated steady-state ERIS study results, NRIS deliverability study results as well as short circuit study results based on the final Phase 3 study models with J807, J819 and J821 removed due to post-GIA withdrawals. Voltage analysis was not rerun as there was no voltage constraint identified in the Phase 3 SIS report. PSSE dynamic stability study was not performed as there were no Non-P6 PSSE stability issues identified in the Phase 3 SIS report. There was one PSCAD stability issue identified for J870/J871 in Phase 3 SIS report, which would not change after the removal of J807, J819 and J821. Therefore, PSCAD study was also not performed. Changes to the Phase 3 ERIS and NRIS Network Upgrades were identified.

Due to a change of one NRIS network upgrade project, the short circuit study was rerun and included in this report. The original short circuit restudy report issued on 07/06/2020 assumed that Hillman transformer upgrade was no longer needed which is incorrect.

The final ERIS and NRIS network upgrades are listed in Section 6 with detailed cost allocation information. The total cost of Network Upgrades required for each generator interconnection request is listed in Table 1-1 below. The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of this report date.

Table 1-1 – Total Cost of Network Upgrades for MISO DPP 2017 August Wisconsin Phase 3 Generator Interconnection Requests

MISO Queue #	Requested MW	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Cost of Network Upgrades (Exclude TOIF & Affected Systems) (\$)	M2 Received (\$) ¹	M3 Received (\$) ²	M4 Received (\$) ⁷
		Steady - State Thermal & Voltage	Dynamic and EMT ⁸ Stability	Short Circuit	Affected System		TO Network Upgrades	TO-Owned Direct Assigned (TOIF)					
[a]	[b]	[c]	[d]	[e]	[f]	[g]	[h]	[i]	[j]	[k]=[c]+[d]+[e]+[g]+[h]	[l] = \$4000 x [b]	[m]= (10%of [k])-[l]	[n]= (20%of [k])-[l] –[m]
J732	561.5	3,898,500	0	440,000	0	0	17,565,722	839,736	0	21,904,222	2,246,000	63,168	2,003,676
J818	149	0	0	440,000	220,000	4,696,165	1,847,766	358,000	0	6,983,931	596,000	0	732,786
J831	40	0	0	0	0	0	0	0	0	0	160,000	0	0
J850	250	0	0	888,890	440,000	9,440,036	7,430,619	246,888	0	17,759,546	1,000,000	1,327,714	1,070,007
J855	100.8	0	0	253,877	182,207	0	8,005,945	241,199	0	8,259,822	403,200	470,823	759,426
J864	49.98	29,753	0	586,025	1,590,982	0	621,293	235,805	0	1,237,071	200,000	0	0
J870	200	640,222	7,776,887	337,462	359,140	0	5,371,900	161,161	0	14,126,470	800,000	0	712,274
J871	100	320,082	3,888,444	168,731	179,570	0	2,685,950	80,581	0	7,063,207	400,000	0	361,010
J878	200	0	0	761,110	0	22,948,305	1,996,777	345,477	0	25,706,193	800,000	1,786,568	2,407,796
J886	150	0	0	0	0	0	1,727,747	757,646	0	1,727,747	600,000	0	0
J947	200	0	0	633,906	1,868,101	2,723,059	7,826,076	236,897	0	11,183,041	800,000	951,616	352,216
Total (\$)	-	4,888,557	11,665,331	4,510,000	4,840,000	39,807,566	55,079,795	3,503,390	0	115,951,249	8,005,200	4,599,889	8,399,191

¹ M2: Milestone Payment dollars received by MISO

² M3 = (10% of NU)-M2

³ M4 = (20% of NU)-M2-M3

⁴ ATC projects are estimated in ISD dollars with a 2.5% annual escalation rate and include a 10% contingency with one exception. J818 TOIF estimate has 0% contingency since it is based on the remaining net book value of the assets.

⁵ For this table, two third of the Interconnection Facilities of the new J870/J871 substation is allocated to J870 and one third of the cost is allocated to J871.

⁶ Transmission Owner shall also collect from Interconnection Customer a tax gross-up amount on the payments made to Transmission Owner using the Transmission Owner rate in effect at the time the payment is received from Interconnection Customer. The current Transmission Owner tax gross-up rate is 12.848%.

⁷ TOIF: Non-reimbursable Transmission Owner Interconnection Facilities to which tax gross-up amount must also be applied.

⁸ EMT stands for electromagnetic transient.

2.0 ERIS STEADY-STATE RESTUDY RESULTS

The final MISO DPP 2017 August Wisconsin Area Phase 3 ERIS study cases were modified by removing J807, J819 and J821. All units in MISO Classic excluding the units in the current DPP cycle were proportionally dispatched to make up the generation reduction. Scheduled firm transfers are ignored in this dispatch methodology. No changes were made to the bench cases.

The incremental impact of the proposed generator interconnection on transmission facilities was evaluated by comparing steady state power flows and voltages between benchmark cases and study cases. Post-contingency cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment enabled, and switched shunt adjustment enabled. Same NERC TPL Category P contingencies that were used in the Phase 3 SIS were studied.

2.1 2022 Summer

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2022 Summer Peak study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2022 Summer Peak MISO Injection Constraints that require Network Upgrades is presented in Table 2.1-1.

Table 2.1-1 – 2022 Summer Peak Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator (s)
2022 Summer Peak	Spring Green – Wyoming Valley 138 kV line	ATC	P1.2	J870 J871
	J870J871 POI – Wyoming Valley 138 kV line	ATC	P1.2	J870 J871
	J870J871 POI – Eden 138 kV line	ATC	P1.2, P2.1	J864 J870 J871
	Stone Lake 345/161 kV transformer	XCEL	P1.2, P1.3, P2.3	J732

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

2.2 2022 Shoulder

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2022 Shoulder study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2022 Shoulder MISO Injection Constraints that require Network Upgrades is presented in Table 2.2-1.

Table 2.2-1 – 2022 Shoulder Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2022 Shoulder	Spring Green – Wyoming Valley 138 kV line	ATC	P1.2, P2.3 EHV	J870 J871
	J870J871 POI – Wyoming Valley 138 kV line	ATC	P1.2, P2.3 EHV	J870 J871

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

2.3 Network Upgrades Identified

Based on the steady-state analyses, the worst loading of each facility under "No Load Loss Allowed" NERC TPL Planning Events that meets MISO Injection Constraint criteria is shown in Table 2.3-1. Good faith cost estimates of the ERIS Network Upgrades identified in the steady-state analysis for the 2022 scenarios are listed in Table 2.3-2. Detailed cost allocations are provided in Section 6.

Table 2.3-1 – ERIS Network Upgrades Identified to Address MISO Steady-State Injection Constraints

Steady-State Injection Constraint	Responsible Generator(s)	Facility Owner	Study Case	Emergency Rating (MVA)	Worst Loading (%)	"No Load Loss Allowed" NERC TPL Planning Event	Network Upgrades
Spring Green – Wyoming Valley 138 kV line	J870 J871	ATC	22SUM	202	140.69		Spring Green – J870J871 POI 138 kV, uprate
J870J871 POI – Wyoming Valley 138 kV line	J870 J871	ATC	22SUM	202	142.37		
J870J871 POI – Eden 138 kV line	J864 J870 J871	ATC	22SUM	202	143.53		Eden – J870J871 POI 138 kV, uprate
Stone Lake 345/161 kV transformer	J732	XCEL	22SUM	386.4	101.33		Stone Lake SS, upgrade transformer

Table 2.3-2 – ERIS Network Upgrades and Cost Estimates

Steady-State Injection Constraint	Facility Owner	Network Upgrade	Cost (\$) ^{1,2}
Spring Green – Wyoming Valley 138 kV line	ATC	Spring Green – J870J871 POI 138 kV, uprate	422,964
J870J871 POI – Wyoming Valley 138 kV line	ATC		
J870J871 POI – Eden 138 kV line	ATC	Eden – J870J871 POI 138 kV, uprate	567,093
Stone Lake 345/161 kV transformer	XCEL	Stone Lake SS, upgrade transformer	3,898,500 ³

¹ All Network Upgrades were estimated on the generator ISD dollars.

² ATC Network Upgrades included a 10% contingency.

³ No contingency was included for the Stone Lake transformer upgrade project according to Xcel Energy.

3.0 DELIVERABILITY RESTUDY RESULTS

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up.

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at

https://cdn.misoenergy.org/Generator_Deliverability_Study_Methodology108139.pdf

3.1 Study Summary and Network Upgrades Identified

The final MISO DPP 2017 August Wisconsin Area Phase 3 NRIS study case was modified by removing J807, J819 and J821. Table 3.1-1 below lists the deliverability summary results with the updated ERIS upgrades included in the NRIS analysis. Minimum NR Deliverable is the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew. Maximum NR Deliverable is the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction.

Table 3.1-1 – NRIS Analysis Summary

MISO Queue #	Area	NR Tested	Minimum NR Deliverable (MW)	Maximum NR Deliverable (MW)
J732	ATC	527.8	527.8	527.8
J818	ATC	149	0	0
J831	ATC	40	40	40
J850	ATC	250	0	127.62
J855	ATC	100.8	100.8	100.8
J864	ATC	49.98	49.98	49.98
J870	ATC	200	200	200
J871	ATC	100	100	100
J878	ATC	200	83.24	102.1
J886	ATC	150	150	150
J947	ATC	200	119.52	119.52

Table 3.1-2 below lists all of the NRIS constraints from the deliverability restudy and the identified NRIS Network Upgrades. Both ERIS Network Upgrades and NRIS Network Upgrades must be made for 100% NRIS, i.e. fully deliverable. Please note, if a NRIS Network Upgrade entirely or partially changes the scope of an ERIS Network Upgrade, only the cost difference between the NRIS upgrade and the ERIS upgrade will be eligible for NRIS Network Upgrade cost allocation. Detailed NRIS Network Upgrade cost allocation calculations are provided in Section 6.

Table 3.1-2 – Network Upgrades Needed to Address MISO Identified NRIS Steady-State Injection Constraints

NRIS Thermal Constraints	Required NRIS Network Upgrades	Mitigation Type	Responsible Generators	NRIS Network Upgrade Cost Estimates (\$) [a]	ERIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [b]	Cost Used for NRIS Network Upgrade Allocation (\$) [c] = [a] - [b]
694105 ROR BUS3 138 699047 ROR 138 138 Z	Rock River SS, bus switch upgrade	NR	J947	95,737	\$0	95,737
699086 ELK 138 138 699085 NLG 138 138 1	Elkhorn – North Lake Geneva 138 kV, uprate	NR	J850	45,400	\$0	45,400
699031 HLM 138 138 698003 HLM 69 69.0 1	Hillman SS, upgrade transformer	NR	J947	2,627,322	\$0	2,627,322
699293 COONEY 138 699470 SUMMIT3 138 1	Cooney – Summit 138 kV line, rebuild (MTEP19 Target A project)	NR	No Allocation	A10	\$0	No Allocation
699410 PARIS WE 138 699982 RAYMOND 138 1 699982 RAYMOND 138 699442 ST MARTIN B3 138 1 699409 PARIS B5 138 693647 BERRYVILLE 138 1	Paris SS, new 345/138 kV transformer	NR	J850, J878	27,410,780	\$0	27,410,780
699141 TOWNLINE 138 699897 BASSCRK 138 1	Bass Creek – Town Line Road 138 kV, reconductor	NR	J850	4,408,401	\$0	4,408,401
698090 BOL 138 138 699086 ELK 138 138 1	Bristol – Elkhorn 138 kV, uprate	NR	J850	243,377	\$0	243,377
699360 NLK GV T 138 699267 BRLGTN1 138 1	North Lake Geneva Tap – Burlington 138 kV, uprate	NR	J850	280,383	\$0	280,383
699283 CONCRD 4 138 699293 COONEY 138 1	Concord – Cooney 138 kV reconductor	NR	J818	4,696,165	\$0	4,696,165

3.2 Per Project Summary

3.2.1 J732

J732 Deliverable (NRIS) Amount in 2022 Case: 363.63 MW (Conditional on ERIS upgrades and case assumptions)	527.8 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		527.8		527.8		J732			

3.2.2 J818

J818 Deliverable (NRIS) Amount in 2022 Case: 59.25MW (Conditional on ERIS upgrades and case assumptions)	149 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
699293 COONEY 138 699470 SUMMIT3 138 1	210.02	0	210.02	0	0.2074	J818	\$4,696,165	\$4,696,165	Cooney - Summit 138 kV line, rebuild Concord - Cooney 138 kV reconductor
		149		149					

3.2.3 J831

J831 Deliverable (NRIS) Amount in 2022 Case: 40 MW (Conditional on ERIS upgrades and case assumptions)	40 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		40		40					

3.2.4 J850

J850 Deliverable (NRIS) Amount in 2022 Case: 0 MW (Conditional on ERIS upgrades and case assumptions)	250 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699410 PARIS WE 138 699982 RAYMOND 138 1	753.53	0	122.38	127.62	0.095	J850, J878	\$4,462,475	\$27,410,780	Paris SS, new 345/138 kV transformer	
699982 RAYMOND 138 699442 ST MARTIN B3 138 1	624.34	0	101.38	148.62	0.095	J850, J878	\$4,462,475	\$27,410,780	Paris SS, new 345/138 kV transformer	
699409 PARIS B5 138 693647 BERRYVILLE 138 1	342.98	0	56.08	193.92	0.107	J850, J878	\$4,462,475	\$27,410,780	Paris SS, new 345/138 kV transformer	
698090 BOL 138 138 699086 ELK 138 138 1	106.7	143.3	106.7	143.3	0.349	J850	\$243,377	\$243,377	Bristol – Elkhorn 138 kV, uprate	
699141 TOWNLINE 138 699897 BASSCRK 138 1	90.73	159.27	90.73	103.77	0.067	J850	\$4,408,401	\$4,408,401	Bass Creek – Town Line Road 138 kV, reconductor	
699360 NLK GV T 138 699267 BRLGTN1 138 1	61.78	188.22	61.78	188.22	0.148	J850	\$280,383	\$280,383	North Lake Geneva Tap – Burlington 138 kV, uprate	
699086 ELK 138 138 699085 NLG 138 138 1	34.5	215.5	34.5	215.5	0.299	J850	\$45,400	\$45,400	Elkhorn – North Lake Geneva 138 kV, uprate	
		250		250						

3.2.5 J855

J855 Deliverable (NRIS) Amount in 2022 Case: 100.8 MW (Conditional on ERIS upgrades and case assumptions)	100.8 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
		100.8		100.8						

3.2.6 J864

J864 Deliverable (NRIS) Amount in 2022 Case: 49.98 MW (Conditional on ERS upgrades and case assumptions)	49.98 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		49.98		49.98					

3.2.7 J870

J870_871 Deliverable (NRIS) Amount in 2022 Case: 189.63 MW (Conditional on ERS upgrades and case assumptions)	200 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		200		200					

3.2.8 J871

J870_871 Deliverable (NRIS) Amount in 2022 Case: 89.63 MW (Conditional on ERS upgrades and case assumptions)	100 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		100		100					

3.2.9 J878

J878 Deliverable (NRIS) Amount in 2022 Case: 70.59 MW (Conditional on ERIIS upgrades and case assumptions)								200 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)								Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699410	PARIS	WE	138	699982	RAYMOND	138	1	116.76	83.24	97.9	102.1	0.615	J850, J878	\$22,948,305	\$27,410,780	Paris SS, new 345/138 kV transformer	
699982	RAYMOND		138	699442	ST MARTIN B3	138	1	96.82	103.18	81.1	118.9	0.615	J850, J878	\$22,948,305	\$27,410,780	Paris SS, new 345/138 kV transformer	
699409	PARIS	B5	138	693647	BERRYVILLE	138	1	53.63	146.37	44.86	155.14	0.682	J850, J878	\$22,948,305	\$27,410,780	Paris SS, new 345/138 kV transformer	
									200		200						

3.2.10 J886

J886 Deliverable (NRIS) Amount in 2022 Case: 150 MW (Conditional on ERIS upgrades and case assumptions)	150 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
		150		150					

3.2.11 J947

J947 Deliverable (NRIS) Amount in 2022 Case: 0 MW (Conditional on ERIS upgrades and case assumptions)	200 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
694105 ROR BUS3 138 699047 ROR 138 138 Z	80.48	119.52	80.48	119.52	0.0729	J947	\$95,737	\$95,737	Rock River SS, bus switch upgrade
699031 HLM 138 138 698003 HLM 69 69.0 1	15.42	184.58	15.42	184.58	0.1684	J947	\$2,627,322	\$2,627,322	Hillman SS, upgrade transformer
		200		200					

4.0 SHORT CIRCUIT RESTUDY RESULTS

Short circuit studies were re-performed to incorporate J807, J819 and J821 post GIA withdrawals, ERIS and NRIS network upgrade changes due to the withdrawals, proposed J870/J871 project design changes as well as a main step-up transformer modeling update for J855.

Based on results in Section 2 and Section 3, the following four network upgrades that were identified in the Phase 3 SIS report are no longer needed due to the three generator withdrawals.

- North Monroe - Bass Creek 138 kV, uprate
- Darlington - North Monroe 138 kV, uprate
- South Monroe – Browntown 69kV, rebuild
- Rock River SS, X-74 terminal upgrade

The DPP-17-Aug Phase 3 protection study CAPE models were updated accordingly, and the short circuit study was re-performed according to ATC short circuit analysis methodology. Short circuit analysis results for facilities under MISO Functional Control are described in this section. Short circuit analysis results for facilities that are not under MISO functional control are described in Section 5 as they are considered as Affected System.

4.1 Short Circuit Study Results

Maximum and minimum fault duty was calculated at the POI for each generator request except for J831 and results are summarized in Table 4.1-1, Table 4.1-2, Table 4.1-3 and Table 4.1-4. J831 is only an NRIS increase request which will not increase short circuit values and so no analysis was performed.

Table 4.1-1 – Outage Assumptions for Minimum Fault Duty Calculations

MISO Queue #	Contingency
J732	J732 POI-ARROWHEAD J732 GENERATION
J818	JEFFERSON-ROCKDALE 138 J818 GENERATION
J850	ROCK RIVER BUS TIE 1-2 138 J850 GENERATION
J855	NELSON DEWEY T91 138-161 XFMR J855 GENERATION
J864	LONE ROCK PHASE SHIFTING XFMR FULL RAISE LONE ROCK-SPRING GREEN 69 J864 GENERATION
J870	EDEN-HILL VALLEY 138 J870 GENERATION

J871	EDEN-HILL VALLEY 138 J871 GENERATION
J878	PARIS-BERRYVILLE 138 PARIS GENERATION & GSUs
J886	KEWAUNEE T20 138-345 XFMR J886 GENERATION
J947	J947 POI-NELSON DEWEY J947 GENERATION

Table 4.1-2 – Maximum and Minimum Fault Duty at POIs

Location	Maximum Fault Duty (Amps)		Minimum Fault Duty (Amps)	
	Single-Phase	Three-Phase	Single-Phase	Three-Phase
J732 POI without J732	4,493	4,950	1,801	2,208
J732 POI with J732	9,841	8,867	7,050	6,126
J818 POI without J818	10,020	13,654	5,518	7,687
J818 POI with J818	12,648	14,087	7,902	8,120
J850 POI without J850	6,065	8,798	4,114	5,857
J850 POI with J850	9,207	9,941	7,036	6,999
J855 POI without J855	12,810	14,067	12,320	13,316
J855 POI with J855	14,288	14,530	13,776	13,779
J864 POI without J864	3,298	4,791	1,154	1,769
J864 POI with J864	6,062	5,231	2,945	2,213
J870 POI without J870	10,036	10,236	4,236	3,555
J870 POI with J870	11,030	10,607	4,926	4,061
J871 POI without J871	10,036	10,236	4,236	3,555
J871 POI with J871	11,030	10,607	4,926	4,061
J878 POI without J878	31,000	30,170	16,732	18,471
J878 POI with J878	33,231	31,804	19,221	20,105
J886 POI without J886	19,687	18,712	11,261	11,054
J886 POI with J886	20,809	18,997	12,482	11,358
J947 POI without J947	4,312	5,360	1,766	2,020
J947 POI with J947	6,941	6,519	3,905	3,179

Table 4.1-3 – Maximum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three-Phase Thevenin X/R Ratio	Single-Phase Thevenin X/R Ratio	Three-Phase ANSI 37.010 X/R Ratio	Single-Phase ANSI 37.010 X/R Ratio
J732 POI without J732	3.30992 + j40.1085	3.32323 + j40.1079	6.83484 + j52.0968	12.1	9.8	12.5	10.5
J732 POI with J732	1.13905 + j22.4346	1.15375 + j22.5697	1.50240 + j15.5999	19.7	16	30.4	21
J818 POI without J818	0.75083 + j5.78868	0.76613 + j5.82268	3.40944 + j11.7110	7.7	4.7	8.2	4.9
J818 POI with J818	0.70694 + j5.61302	0.721078 + j5.64506	1.42798 + j7.40642	7.9	6.5	9.5	10
J850 POI without J850	1.12439 + j8.98306	1.16324 + j8.98926	4.87222 + j20.7369	8	5.4	8.5	5.6
J850 POI with J850	1.12439 + j8.98306	1.16324 + j8.98926	1.46318 + j11.0910	8	7.7	8.5	10.9
J855 POI without J855	0.59953 + j6.20423	0.60711 + j6.20332	1.27550 + j7.64391	10.3	8.1	14.5	11.2
J855 POI with J855	0.59953 + j6.20423	0.60711 + j6.20332	0.89144 + j6.29900	10.4	8.9	14.5	14
J864 POI without J864	2.27985 + j8.13102	2.28162 + j8.13076	4.94258 + j18.8838	3.6	3.7	4.8	5.6
J864 POI with J864	2.27985 + j8.13102	2.28162 + j8.13076	0.52738 + j4.75586	3.6	4.1	4.8	5.6
J870 POI without J870	1.01364 + j8.25803	1.02238 + j8.25703	1.52374 + j9.01171	8.1	7.2	10.1	8.8
J870 POI with J870	1.01364 + j8.25803	1.02238 + j8.25703	1.28996 + j7.64870	8.3	7.3	10.2	8.8
J871 POI without J871	1.01364 + j8.25803	1.02238 + j8.25703	1.52374 + j9.01171	8.1	7.2	10.1	8.8
J871 POI with J871	1.01364 + j8.25803	1.02238 + j8.25703	1.28996 + j7.64870	8.3	7.3	10.2	8.8
J878 POI without J878	0.18006 + j2.63484	0.18668 + j2.73190	0.17927 + j2.32345	14.6	14.1	25.5	24.7
J878 POI with J878	0.18006 + j2.63484	0.18668 + j2.73190	0.15314 + j2.11781	14.6	14.4	25.5	25.5
J886 POI without J886	0.24554 + j4.25210	0.28628 + j4.24618	0.31939 + j3.58761	17.3	14.2	32.2	24
J886 POI with J886	0.24560 + j4.25261	0.28635 + j4.24668	0.31990 + j3.26786	17.3	13.8	32.2	23.5
J947 POI without J947	2.66925 + j14.8303	2.67313 + j14.8301	5.76453 + j24.7415	5.6	4.9	5.9	5.3
J947 POI with J947	2.67058 + j14.8338	2.67446 + j14.8337	1.58526 + j 11.8216	5.6	6	5.9	6.7

Table 4.1-4 – Minimum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three- Phase Thevenin X/R Ratio	Single- Phase Thevenin X/R Ratio	Three- Phase ANSI 37.010 X/R Ratio	Single- Phase ANSI 37.010 X/R Ratio
J732 POI without J732	8.55825 + j89.8220	8.62507 + j89.8172	22.7468 + j149.721	10.5	8.2	11.3	9.5
J732 POI with J732	1.33733 + j32.4878	1.36279 + j32.7722	1.75493 + j19.3830	24.3	19	38	25.1
J818 POI without J818	1.43162 + j10.2653	1.44554 + j10.3863	6.36338 + j21.6701	7.2	4.6	7.6	4.8
J818 POI with J818	1.28787 + j9.72662	1.29883 + j9.83510	1.62126 + j10.3917	7.6	7.1	9.8	11
J850 POI without J850	1.85314 + j13.4770	1.86726 + j13.4999	6.77543 + j30.1142	7.3	5.4	6.9	5.8
J850 POI with J850	1.85314 + j13.4770	1.86726 + j13.4999	1.43523 + j13.2626	7.3	7.8	6.9	9.1
J855 POI without J855	0.60439 + j6.59016	0.61309 + j6.58925	1.27886 + j7.74869	10.9	8.4	14.5	11.3
J855 POI with J855	0.60439 + j6.59016	0.61309 + j6.58925	0.89022 + j6.36947	10.9	9.3	15.4	14.6
J864 POI without J864	6.85461 + j21.4480	6.85482 + j21.4481	15.4239 + j56.3985	3.1	3.4	3.2	2.9
J864 POI with J864	6.85461 + j21.4480	6.85482 + j21.4481	0.47447 + j5.68615	3.1	3.4	3.2	3.5
J870 POI without J870	4.89094 + j25.1092	4.90593 + j25.1074	1.58545 + j13.6118	5.1	5.6	5.6	6.5
J870 POI with J870	4.89094 + j25.1092	4.90593 + j25.1074	1.36168 + j10.7293	5.1	5.5	1.3	-1.7
J871 POI without J871	4.89094 + j25.1092	4.90593 + j25.1074	1.58545 + j13.6118	5.1	5.6	5.6	6.5
J871 POI with J871	4.89094 + j25.1092	4.90593 + j25.1074	1.36168 + j10.7293	5.1	5.5	1.3	-1.7
J878 POI without J878	0.31463 + j4.30230	0.31982 + j4.30852	0.60308 + j5.61884	13.7	11.5	25.5	19
J878 POI with J878	0.31463 + j4.30230	0.31982 + j4.30852	0.41468 + j4.55298	13.7	12.5	25.5	22.9
J886 POI without J886	0.54297 + j7.18057	0.57905 + j7.17389	0.53448 + j6.78720	13.2	12.8	31.7	25.9
J886 POI with J886	0.54318 + j7.18201	0.57928 + j7.17532	0.51098 + j5.66749	13.2	12.3	31.7	24.9
J947 POI without J947	7.98446 + j38.8964	7.99952 + j38.8959	12.2910 + j54.0878	4.9	4.7	5.1	5
J947 POI with J947	7.99489 + j38.9208	8.00998 + j38.9202	1.58124 + j15.8836	4.9	5.3	5.1	5.7

ATC breaker duty analysis results for over duty breakers are summarized in Table 4.1-5 and Table 4.1-6.

Table 4.1-5 – ATC Over Duty Breakers Found in SLG Analysis

Breaker Name	Breaker Rating	Base Case SLG Margin	Study Case SLG Margin
None	N/A	N/A	N/A

Table 4.1-6 – ATC Over Duty Breakers Found in 3PG Analysis

Breaker Name	Breaker Rating	Base Case 3PG Margin	Study Case 3PG Margin
None	N/A	N/A	N/A

Please note, no breaker duty analysis was performed for Affected System equipment. Appendix M provides all substation buses included in ATC protection CAPE models that show a 1% increase in SLG or 3PG fault current. Any non-ATC equipment requiring a change would be considered Affected System.

Ground fault analysis results are summarized in Table 4.1-7.

Table 4.1-7 – Buses with 10% or More Fault Current Increase in SLG Analysis

SUBSTATION	Facility Owner	BUS	kV	BEFORE (AMPS)	AFTER (AMPS)	% Change
EDEN	ALTE	801600	69	7780.4	10110.4	29.95
EDEN	ALTE	903400	138	10927.9	13809.6	26.37
SOUTHWEST DELAVAN	ALTE	894600	138	5590	6892.9	23.31
WEST DARIEN	ALTE	809300	138	5487.4	7254.5	32.20
BRISTOL (ALTE)	ATC	809000	138	6058.1	6948.3	14.69
BRISTOL (ALTE)	ATC	809001	138	6058.1	6948.3	14.69
CRAWFISH RIVER	ATC	937500	138	9196.5	10875.7	18.26
CRAWFISH RIVER	ATC	937501	138	9196.5	10875.7	18.26
DODGEVILLE	ATC	802200	69	2916.6	3314	13.63
FALCON	ATC	340300	138	4418.9	5022.3	13.65
HILL VALLEY	ATC	386300	345	6541.3	7413.9	13.34
HILLMAN	ATC	800300	69	4026.1	6147.1	52.68
HILLMAN	ATC	903100	138	4166.3	5429.5	30.32
LONE ROCK	ATC	812300	69	3165.3	5851	84.85
LONE ROCK	ATC	812400	69	2359	3418.6	44.92
NELSON DEWEY	ATC	902000	138	7726.2	8912.3	15.35
NELSON DEWEY	ATC	902001	138	7726.2	8912.3	15.35
NELSON DEWEY	ATC	901000	161	6999.9	7765.1	10.93
SPRING GREEN	ATC	812700	69	6429.2	7284.9	13.31

SPRING GREEN	ATC	812701	69	6429.2	7284.9	13.31
SPRING GREEN	ATC	911400	138	4664.8	5216.8	11.83
ARROWHEAD	ATC / MP	944900	345	5385	8023.5	49.00
ARROWHEAD	ATC / MP	944800	230	10244.8	12779.7	24.74
ARROWHEAD	ATC / MP	608615	230	12421.9	14394.8	15.88
STONE LAKE	ATC / XCEL	945000	345	4588.5	5657.7	23.30
STONE LAKE	ATC / XCEL	602017	161	8839.1	10299.1	16.52
REC LAPRAIRIE	REC	807200	138	9345.9	10875	16.36
RICHLAND CENTER	RICHLAND CENTER	812600	69	2060.7	2291.3	11.19
BERRYVILLE	WEC	364700	138	12932.3	14854.8	14.87
BERRYVILLE	WEC	364701	138	12932.3	14854.8	14.87
JEFFERSON	WEC	888400	138	10019.1	12648	26.24
JEFFERSON	WEC	934000	138	10019.1	12648	26.24
PARIS	WEC	940200	138	20170.6	33231	64.75
PARIS	WEC	940900	138	20170.6	33231	64.75
PARIS	WEC	941000	138	20170.6	33231	64.75

4.2 Short Circuit Network Upgrades Identified

Short circuit Network Upgrades are summarized in Table 4.2-1. Details on cost estimating/allocation methodology and cost allocations are provided in Section 6.

Table 4.2-1 – Substation Grounding Grid Upgrades

SUBSTATION	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)
WEST DARIEN	ALTE	ATC	220,000
EDEN	ALTE	ATC	220,000
SOUTHWEST DELAVAN	ALTE	ATC	220,000
LONE ROCK	ATC	ALTE	220,000
HILLMAN	ATC	ALTE	220,000
HILL VALLEY	ATC	-	220,000
CRAWFISH RIVER	ATC	-	220,000
NELSON DEWEY	ATC	ALTE	220,000
BRISTOL (ALTE)	ATC	ALTE	220,000
FALCON	ATC	-	220,000
DODGEVILLE	ATC	ALTE	220,000
SPRING GREEN	ATC	ALTE	220,000
ARROWHEAD	ATC / MP	-	220,000
STONE LAKE	ATC / XCEL	WPS	220,000
REC LAPRAIRIE	REC	ATC	220,000
RICHLAND CENTER	RICHLAND CENTER	ATC	220,000
PARIS	WEC	ATC	550,000
JEFFERSON	WEC	ATC	220,000
BERRYVILLE	WEC	ATC	220,000

5.0 AFFECTED SYSTEM SHORT CIRCUIT RESTUDY RESULTS

Some affected system facilities that are not under MISO functional control are modeled in the ATC Protection CAPE models and were evaluated in the short circuit analysis.

5.1 Short Circuit Study Results

Affected system ground fault analysis results are summarized in Table 5.1-1.

Table 5.1-1 – Affected System Buses with 10% or More Fault Current Increase in SLG Analysis

SUBSTATION	Facility Owner	BUS	kV	BEFORE (AMPS)	AFTER (AMPS)	% Change
POTOSI	ALTE	903000	138	4164.5	6941	66.67
MCGREGOR (ALTE)	ALTE	800400	69	3511.5	5019.2	42.94
PIONEER(ALTE)	ALTE	800500	69	3256.5	4508.3	38.44
AVOCA	ALTE	812100	69	2080.6	2791.7	34.18
ELMO	ALTE	800900	69	2659.8	3230.8	21.47
BELMONT	ALTE	801400	69	2904.8	3476.3	19.67
RICHLAND	ALTE	812500	69	2013.2	2317.6	15.12
LANDS END	ALTE	802300	69	2863.4	3292.1	14.97
WYOMING VALLEY	ALTE	911500	138	4468.7	5096.8	14.06
BLUE RIVER	ALTE	811800	69	1837.2	2079.9	13.21
DELANVAN	ALTE	809100	138	5190.2	5831.2	12.35
PINE RIVER (ALTE)	ALTE	812200	69	2037.7	2288.5	12.31
BENTON	ALTE	801000	69	2121.5	2357.7	11.13
REWEY	ALTE	801500	69	3027.4	3361.5	11.04
CUBA CITY	CUBA CITY MUNI	801300	69	2311.5	2658.1	14.99
CUBA CITY SOUTH	CUBA CITY MUNI	801350	69	2185	2458.3	12.51
CUBA CITY SOUTH	CUBA CITY MUNI	801351	69	2008.4	2236.9	11.38
DPC PLATTEVILLE	DPC	800600	69	3490.6	4971.1	42.41
DPC PLATTEVILLE	DPC	680071	69	3110.5	4232.3	36.06
MUSCODA INDUSTRIALS	MUSCODA MUNI	383500	69	1998.1	2524.8	26.36
MUSCODA	MUSCODA MUNI	811900	69	1986.8	2497.5	25.70
REC BRADFORD	REC	809400	138	6039.5	9149.7	51.50
BREWER	RICHLAND CENTER	812250	69	2038.2	2288.4	12.28
FORT ATKINSON	WEC	930500	138	5845.2	6662.8	13.99
FORT ATKINSON	WEC	929800	138	5653.5	6417.4	13.51

Please note, no breaker duty analysis was performed for Affected System equipment. Appendix M provides all substation buses included in ATC protection CAPE models that

show a 1% increase in SLG or 3PG fault current. Any non-ATC equipment requiring a change would be considered Affected System.

5.2 Network Upgrades Identified in the Short Circuit Analysis

Affected short circuit Network Upgrades are summarized in Table 5.2-1. Details on cost estimating/allocation methodology and cost allocations are provided in Section 6.

Table 5.2-1 – Network Upgrade Required for Affected System Facilities

SUBSTATION	Facility Owner	Total Cost (\$)
POTOSI	ALTE	220,000
MCGREGOR (ALTE)	ALTE	220,000
PIONEER(ALTE)	ALTE	220,000
AVOCA	ALTE	220,000
ELMO	ALTE	220,000
BELMONT	ALTE	220,000
RICHLAND	ALTE	220,000
LANDS END	ALTE	220,000
WYOMING VALLEY	ALTE	220,000
BLUE RIVER	ALTE	220,000
DELAVAN	ALTE	220,000
PINE RIVER (ALTE)	ALTE	220,000
BENTON	ALTE	220,000
REWEY	ALTE	220,000
CUBA CITY	CUBA CITY MUNI	220,000
CUBA CITY SOUTH	CUBA CITY MUNI	220,000
DPC PLATTEVILLE	DPC	220,000
MUSCODA INDUSTRIAL S	MUSCODA MUNI	220,000
MUSCODA	MUSCODA MUNI	220,000
REC BRADFORD	REC	220,000
BREWER	RICHLAND CENTER	220,000
FORT ATKINSON	WEC	220,000

6.0 UPDATED COST ALLOCATION TABLES

6.1 Cost Allocation Methodology for Thermal Network Upgrades

The costs of Network Upgrades (NU) for a set of generation projects (one or more subgroups or entire group with identified NU) are based off the MW impact of the worst-case scenario for each specific generator project. Basically, whatever the highest MW impact (increasing flow) is for that particular generator where the constraint is identified and requires NU is how it should be calculated.

Constraints which are mitigated by one or a subset of NU are identified. The highest MW contribution on these constraints from each generating facility is calculated in the MISO DPP study models without any Network Upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contribution on constraint)}} \right)$$

6.2 Cost Estimating and Allocation Methodology for Short Circuit Upgrades

For each ATC breaker shown to be loaded above 100% of rating a new breaker will be scoped and the cost of that upgrade will be assigned by to generators based on the MW impact provided from each generator for the worst case loading of the breaker.

For each ATC substation that shows a bus having fault current (SLG) increased by 10% or more, ATC will perform detailed grounding grid analysis if in-house grounding grid modeling is available at the time of the study. For the substations that pass the detailed grounding analysis (i.e. meet the safety criteria), they will be removed from the cost allocation process. For the remaining substations, costs are assigned for ground grid upgrades at that substation.

Only one cost estimate is scoped per substation regardless of the number of buses at that substation that show a 10% or greater increase in fault current. The largest fault current value at a substation in the After Case will determine the ground grid upgrade costs. If the largest fault current is above 20 kA, the upgrades are assigned a cost estimate of \$550,000. If the largest fault current is at or below 20 kA, the upgrades are assigned a cost estimate of \$220,000 for ground grid upgrades only.

Once costs are determined, they are allocated proportionally to study generators that have a greater than 3% of the total of all current queue study generator contributing fault currents under the single line-ground short circuit fault simulation at the identified substation, requiring potential grounding grid upgrades.

This methodology is applied to both ATC and non-ATC facilities modeled in ATC Protection models. For non-ATC facilities, it is interconnection customer's responsibility to work with facility owners to further refine costs, and

implementation of the mitigation projects (if needed). These are placeholder costs until facility studies are performed. Based on those results, costs are adjusted.

6.3 Cost Allocation Tables

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of this final report date. The cost estimate for each network upgrade was provided by the corresponding transmission owning company. Assuming all the remaining Aug-17 cycle generating facilities advance, Table 6.3-1, Table 6.3-2, Table 6.3-3, Table 6.3-4, Table 6.3-5, Table 6.3-6, Table 6.3-7 show how the costs for ERIS steady-state Network Upgrades, EMT stability Network Upgrades, NRIS Network Upgrades and short circuit allocated to responsible generating facilities.

Table 6.3-1 – ERIS Steady-State Network Upgrade Costs Allocated to Each Generation Project

ERIS Steady-State Thermal Injection Constraints	Required ERIS Network Upgrades	Cost Estimates (\$)	Worst MW Impact				Total MW Impact	Cost Allocation %				Cost Allocation (\$)			
			J732	J864	J870	J871		J732	J864	J870	J871	J732	J864	J870	J871
Stone Lake 345/161 kV transformer	Stone Lake SS, upgrade transformer	3,898,500	240.49	-	-	-	240.49	100%	-	-	-	3,898,500	-	-	-
J870J871 POI – Eden 138 kV line	J870J871 POI – Eden 138 kV, uprate	422,964	-	22.7	200	100	322.7	-	7.03%	61.98%	30.99%	-	29,753	262,141	131,070
J870J871 POI – Wyoming Valley 138 kV line	J870J871 POI – Spring Green 138 kV, uprate	567,093	-	-	200	100	300	-	-	66.67%	33.33%	-	-	378,081	189,012
Spring Green – Wyoming Valley 138 kV line			-	-	200	100	300	-	-	66.67%	33.33%				
Total ERIS Steady State Thermal Network Upgrade Cost (\$) Allocated to Each Responsible Generator												3,898,500	29,753	640,222	320,082

Table 6.3-2 – ERIS EMT Stability Network Upgrade Costs Allocated to Each Generation Project

Electromagnetic Transient Instability Identified for the following generators	Required ERIS Network Upgrades	Cost Estimates (\$)	Cost Allocation %		Cost Allocation (\$)	
			J870	J871	J870	J871
J870, J871, J864	Eden - Lands End 69 kV, loop into J870/J871 POI	11,665,331	67%	33%	7,776,887	3,888,444

Table 6.3-3 – NRIS Network Upgrade Costs Allocated to Each Generation Project (Part 1)

Required NRIS Network Upgrades	Responsible Generators	Cost Used for NRIS Network Upgrade Allocation (\$)	MW Impact and % Cost Allocation Details		
Rock River SS, bus switch upgrade	J947	95,737	J947 100%	-	-
Elkhorn – North Lake Geneva 138 kV, uprate	J850	45,400	J850 100%	-	-
Hillman SS, upgrade transformer	J947	2,627,322	J947 100%	-	-
Cooney – Summit 138 kV line, rebuild (MTEP19 Target A project)	No Allocation	No Allocation	-	-	-
Paris SS, new 345/138 kV transformer	J850, J878	27,410,780	J850, 74.32 MW impact, 16.28%	J878, 382.22 MW impact, 83.72%	456.54 total MW impact
Bass Creek – Town Line Road 138 kV, reconductor	J850	4,408,401	J850 100%	-	-
Bristol – Elkhorn 138 kV, uprate	J850	243,377	J850 100%	-	-
North Lake Geneva Tap – Burlington 138 kV, uprate	J850	280,383	J850 100%	-	-
Concord – Cooney 138 kV reconductor	J818	4,696,165	J818 100%	-	-

Table 6.3-4 – NRIS Network Upgrade Costs Allocated to Each Generation Project (Part 2)

Required NRIS Network Upgrades	Cost Used for NRIS Network Upgrade Allocation (\$)	% Cost Allocation				Cost Allocation			
		J818	J850	J878	J947	J818	J850	J878	J947
Rock River SS, bus switch upgrade	95,737	-	-	-	100%	-	-	-	95,737
Elkhorn – North Lake Geneva 138 kV, uprate	45,400	-	100%	-	-	-	45,400	-	-
Hillman SS, upgrade transformer	2,627,322	-	-	-	100%	-	-	-	2,627,322
Cooney – Summit 138 kV line, rebuild (MTEP19 Target A project)	No Allocation	-	-	-	-	-	-	-	-
Paris SS, new 345/138 kV transformer	27,410,780	-	16.28%	83.72%	-	-	4,462,475	22,948,305	-
Bass Creek – Town Line Road 138 kV, reconductor	4,408,401	-	100%	-	-	-	4,408,401	-	-
Bristol – Elkhorn 138 kV, uprate	243,377	-	100%	-	-	-	243,377	-	-
North Lake Geneva Tap – Burlington 138 kV, uprate	280,383	-	100%	-	-	-	280,383	-	-
Concord – Cooney 138 kV reconductor	4,696,165	100%	-	-	-	4,696,165	-	-	-
Total NRIS Network Upgrade Cost (\$) Allocated to Each Responsible Generator						4,696,165	9,440,036	22,948,305	2,723,059

Table 6.3-5 – ERIS Short-Circuit Grounding Grid Network Upgrade Costs Allocated to Each Generation Project
(Part 1a - Substations Under MISO Functional Control)

SUBSTATION	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)	% Cost Allocation								
				J732	J818	J850	J855	J864	J870	J871	J878	J947
WEST DARIEN	ALTE	ATC	220,000	-	-	100%	-	-	-	-	-	-
EDEN	ALTE	ATC	220,000	-	-	-	29.90%	-	41.88%	20.94%	-	7.28%
SOUTHWEST DELAVAN	ALTE	ATC	220,000	-	-	100%	-	-	-	-	-	-
LONE ROCK	ATC	ALTE	220,000	-	-	-	-	100%	-	-	-	-
HILLMAN	ATC	ALTE	220,000	-	-	-	3.81%	-	4.51%	2.25%	-	89.43%
HILL VALLEY	ATC	-	220,000	-	-	-	49.07%	-	31.52%	15.76%	-	3.66%
CRAWFISH RIVER	ATC	-	220,000	-	100%	-	-	-	-	-	-	-
NELSON DEWEY	ATC	ALTE	220,000	-	-	-	10.30%	3.54%	5.18%	2.59%	-	78.39%
BRISTOL (ALTE)	ATC	ALTE	220,000	-	-	100%	-	-	-	-	-	-
FALCON	ATC	-	220,000	-	-	-	-	-	-	-	-	100%
DODGEVILLE	ATC	ALTE	220,000	-	-	-	16.62%	3.43%	47.05%	23.52%	-	9.38%
SPRING GREEN	ATC	ALTE	220,000	-	-	-	5.70%	59.40%	23.27%	11.63%	-	-
ARROWHEAD	ATC / MP	-	220,000	100%	-	-	-	-	-	-	-	-
STONE LAKE	ATC / XCEL	WPS	220,000	100%	-	-	-	-	-	-	-	-
REC LAPRAIRIE	REC	ATC	220,000	-	-	100%	-	-	-	-	-	-
RICHLAND CENTER	RICHLAND CENTER	ATC	220,000	-	-	-	-	100%	-	-	-	-
PARIS	WEC	ATC	550,000	-	-	-	-	-	-	-	100%	-
JEFFERSON	WEC	ATC	220,000	-	100%	-	-	-	-	-	-	-
BERRYVILLE	WEC	ATC	220,000	-	-	4.04%	-	-	-	-	95.96%	-

Table 6.3-6 – ERIIS Short-Circuit Network Upgrade Costs Allocated to Each Generation Project
(Part 1b - Substations Under MISO Functional Control)

SUBSTATION	Facility Owner	Tenants for Joint Use Substations	Total Cost (\$)	Cost Allocation (\$)								
				J732	J818	J850	J855	J864	J870	J871	J878	J947
WEST DARIEN	ALTE	ATC	220,000	-	-	220,000	-	-	-	-	-	-
EDEN	ALTE	ATC	220,000	-	-	-	65,778	-	92,138	46,069	-	16,014
SOUTHWEST DELAVAN	ALTE	ATC	220,000	-	-	220,000	-	-	-	-	-	-
LONE ROCK	ATC	ALTE	220,000	-	-	-	-	220,000	-	-	-	-
HILLMAN	ATC	ALTE	220,000	-	-	-	8,373	-	9,919	4,960	-	196,748
HILL VALLEY	ATC	-	220,000	-	-	-	107,952	-	69,334	34,667	-	8,046
CRAWFISH RIVER	ATC	-	220,000	-	220,000	-	-	-	-	-	-	-
NELSON DEWEY	ATC	ALTE	220,000	-	-	-	22,671	7,793	11,387	5,693	-	172,456
BRISTOL (ALTE)	ATC	ALTE	220,000	-	-	220,000	-	-	-	-	-	-
FALCON	ATC	-	220,000	-	-	-	-	-	-	-	-	220,000
DODGEVILLE	ATC	ALTE	220,000	-	-	-	36,561	7,548	103,499	51,750	-	20,641
SPRING GREEN	ATC	ALTE	220,000	-	-	-	12,541	130,683	51,184	25,592	-	-
ARROWHEAD	ATC / MP	-	220,000	220,000	-	-	-	-	-	-	-	-
STONE LAKE	ATC / XCEL	WPS	220,000	220,000	-	-	-	-	-	-	-	-
REC LAPRAIRIE	REC	ATC	220,000	-	-	220,000	-	-	-	-	-	-
RICHLAND CENTER	RICHLAND CENTER	ATC	220,000	-	-	-	-	220,000	-	-	-	-
PARIS	WEC	ATC	550,000	-	-	-	-	-	-	-	550,000	-
JEFFERSON	WEC	ATC	220,000	-	220,000	-	-	-	-	-	-	-
BERRYVILLE	WEC	ATC	220,000	-	-	8,890	-	-	-	-	211,110	-
Total ERIIS Short Circuit NU Cost (\$) Allocated to Each Generator				440,000	440,000	888,890	253,877	586,025	337,462	168,731	761,110	633,906

Table 6.3-7 – ERIIS Short-Circuit Network Upgrade Costs Allocated to Each Generation Project
(Part 2 - Affected System)

SUBSTATION	Facility Owner	Total Cost (\$)	% Cost Allocation							Cost Allocation (\$)						
			J818	J850	J855	J864	J870	J871	J947	J818	J850	J855	J864	J870	J871	J947
POTOSI	ALTE	220,000	-	-	-	-	-	-	100%	-	-	-	-	-	-	220,000
MCGREGOR (ALTE)	ALTE	220,000	-	-	3.76%	-	4.44%	2.22%	89.57%	-	-	8,281	-	9,773	4,887	197,059
PIONEER(ALTE)	ALTE	220,000	-	-	3.81%	-	4.51%	2.25%	89.42%	-	-	8,393	-	9,917	4,959	196,731
AVOCA	ALTE	220,000	-	-	-	100%	-	-	-	-	-	-	220,000	-	-	-
ELMO	ALTE	220,000	-	-	3.86%	-	4.50%	2.25%	89.39%	-	-	8,495	-	9,901	4,950	196,653
BELMONT	ALTE	220,000	-	-	9.64%	-	12.88%	6.44%	71.04%	-	-	21,205	-	28,344	14,172	156,280
RICHLAND	ALTE	220,000	-	-	-	100%	-	-	-	-	-	-	220,000	-	-	-
LANDS END	ALTE	220,000	-	-	16.35%	3.36%	48.11%	24.06%	8.12%	-	-	35,968	7,401	105,851	52,926	17,855
WYOMING VALLEY	ALTE	220,000	-	-	10.30%	24.21%	43.66%	21.83%	-	-	-	22,668	53,253	96,053	48,026	-
BLUE RIVER	ALTE	220,000	-	-	-	95.60%	-	-	4.40%	-	-	-	210,329	-	-	9,671
DELAVAN	ALTE	220,000	-	100%	-	-	-	-	-	-	220,000	-	-	-	-	-
PINE RIVER (ALTE)	ALTE	220,000	-	-	-	100%	-	-	-	-	-	-	220,000	-	-	-
BENTON	ALTE	220,000	-	-	3.94%	-	4.51%	2.26%	89.29%	-	-	8,669	-	9,927	4,964	196,439
REWEY	ALTE	220,000	-	-	19.54%	-	27.05%	13.53%	39.88%	-	-	42,983	-	59,520	29,760	87,737
CUBA CITY	CUBA CITY MUNI	220,000	-	-	3.89%	-	4.54%	2.27%	89.30%	-	-	8,567	-	9,981	4,991	196,461
CUBA CITY SOUTH	CUBA CITY MUNI	220,000	-	-	3.91%	-	4.53%	2.26%	89.29%	-	-	8,607	-	9,963	4,981	196,449
DPC PLATTEVILLE	DPC	220,000	-	-	3.80%	-	4.50%	2.25%	89.44%	-	-	8,370	-	9,909	4,955	196,766
MUSCODA INDUSTRIALS	MUSCODA MUNI	220,000	-	-	-	100%	-	-	-	-	-	-	220,000	-	-	-
MUSCODA	MUSCODA MUNI	220,000	-	-	-	100%	-	-	-	-	-	-	220,000	-	-	-
REC BRADFORD	REC	220,000	-	100%	-	-	-	-	-	-	220,000	-	-	-	-	-
BREWER	RICHLAND CENTER	220,000	-	-	-	100%	-	-	-	-	-	-	220,000	-	-	-
FORT ATKINSON	WEC	220,000	100%	-	-	-	-	-	-	220,000	-	-	-	-	-	-
Total ERIIS Short Circuit NU Cost (\$) Allocated to Each Generator										220,000	440,000	182,207	1,590,982	359,140	179,570	1,868,101

7.0 UPDATED APPENDIX DOCUMENTS (NOT ATTACHED)

Appendix E – Steady State Power Flow Results

Appendix I – MISO Deliverability Study Results

Appendix M – Short Circuit Data